A Deployed 1-Bit Cross-Modal Hashing System: Browser-Native Multilingual Image Search at 128 Bytes per Image

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Abstract

We present a deployed image–text retrieval system that compresses a frozen vision–language encoder (SigLIP2-So400m) into **1024-bit (128-byte)** binary codes and runs search entirely client-side via Hamming distance, with no retrieval backend. A single Matryoshka hashing head (per-bit BatchNorm + L2, sign straight-through) turns the frozen embedding into nested codes that are independently usable at every prefix length (16–1024 bit). On COCO 5K the deployed 1-bit codes reach **R@10 81.24 (English)** and **R@10 71.04 (Korean)**, recovering most of the float-cosine ceiling (83.58 / 66.26) and far exceeding naive sign hashing without a head (72.62 / 52.18). Against same-feature unsupervised hashing baselines (ITQ, LSH) the learned head wins by a margin that widens sharply at low bit-rates (e.g. $+55$ R@10 at 64-bit). We extend the system to 36 languages (XM3600) and to fully on-device operation: a browser query path (int8 multilingual text encoder) and on-device photo indexing via head-adapted mobile vision encoders that map into the same frozen code space. We report measured phone latency and the runtime (not accuracy) constraints that force fp32-only browser inference. We do not claim to beat float multilingual SoTA (NLLB-CLIP, XM3600 R@10 86.06); our contribution is the deployable, backend-free, portable-index regime and an honest characterization of where it holds and where it does not.

1 Introduction

Frozen-feature deep hashing—training a small head on a fixed foundation-model embedding with a single objective—has recently been shown effective for unimodal retrieval (OrthoHash [1]; HashCoder/CroVCA [2]). Those works establish that BatchNorm-based code balance and a single alignment loss suffice; we claim *no* novelty on that point and our own loss-composition ablations are consistent with theirs (composition and weight-tuning are inert; BatchNorm dominates the code geometry). The open lane, and our focus, is the **cross-modal** + **multilingual** + **on-device** extension: image–text retrieval, non-English text, and browser/phone deployment, together with the failure modes that only appear there. Concretely, this paper contributes (i) a deployed browser-native 1-bit cross-modal search system at 128 B/image (§2); (ii) main retrieval, multilingual, baseline, on-device-cost, and encoder-choice results, all measured (§3); and (iii) analysis hooks (§4–§7) for the main-track companion. Throughout we prioritize measured, reproducible numbers over claims; every table cites the CSV it is computed from.

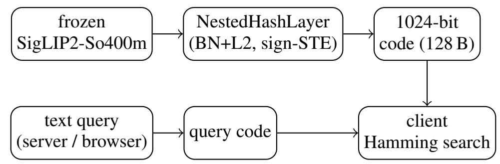

Figure 1: System architecture. Gallery codes (top) are precomputed once; queries (bottom) are encoded by one of two text paths (§2.2) and matched by client-side Hamming distance. No retrieval backend.

2 System

2.1 Overview

The pipeline is a frozen encoder followed by a lightweight hashing head and client-side search (Fig. 1). A frozen SigLIP2-So400m encoder produces a 1152-d embedding e; a *NestedHashLayer* maps e to a 1024-bit code. The head is a single projection ($1152 \to 384 \to 1024$) whose output is sliced to each target length b, then per-bit BatchNorm and L2 normalization are applied, and the sign function (with a straight-through estimator for training) yields the binary code $\mathbf{b} \in \{-1, +1\}^b$. Because shorter codes are strict prefixes of longer ones, a single model serves every bit-rate $16 \to 1024$; the deployed configuration stores the full **1024-bit = 128-byte** code per image and ranks by Hamming distance in the client.

2.2 Two query paths

The gallery is fixed (SigLIP2 image codes), but queries can be encoded two ways. The **server path** runs the native SigLIP2 text tower, used when a (stateless) text-encoding endpoint is available; it gives the headline numbers in §3. The **browser path** runs a small multilingual text encoder exported to int8 ONNX, whose output is mapped into the SigLIP2 code space by an adapted head txt_h'; this path is used for fully offline / private operation and trades accuracy for backend independence (§3.2, §3.5). Both paths emit codes in the same 1024-bit space, so they share one gallery index.

2.3 On-device image indexing

For private, on-device galleries the user's own photos must be encoded without a server. We freeze a mobile vision encoder (e.g. MobileCLIP2) and train an image-side adapter img_h' that maps its embedding into the *same* frozen SigLIP2 1024-bit code space behind the index. The gallery therefore stays

lang	method	R@1	R@5	R@10	mAP@10
EN	float ceiling	52.94	75.72	83.58	62.71
EN	**ours 1-bit**	43.40	71.16	**81.24**	55.15
EN	naive sign	38.42	62.76	72.62	48.95
KO	float ceiling	31.88	55.32	66.26	42.15
KO	**ours 1-bit**	30.68	58.98	**71.04**	42.62
KO	naive sign	21.74	42.14	52.18	30.50

Table 1: COCO 5K, 1024-bit, lowercased eval. *Source: paper/coco_lc_full.csv.*

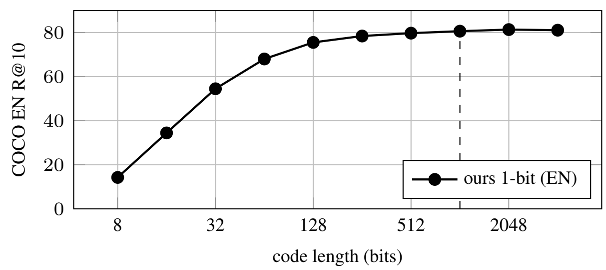

Figure 2: Bit-rate sweep, COCO EN, lowercased. *Source: paper/bits_extreme_lc.csv (lower rows).*

configuration	XM3600 avg-36 R@10	deployable
float ceiling (SigLIP2)	74.46	no
server (ours, 1-bit)	**70.30**	no
offline MiniLM (ours, 1-bit)	53.92	**browser**

Table 2: XM3600 36-language average. Offline beats server on {hi, te, th, mi}. *Source: paper/multiling_server_lc.csv, paper/master_table.csv.*

a single frozen code space regardless of which on-device encoder produced a given entry, and server-encoded and phone-encoded images coexist in one index. §3.5 quantifies the accuracy of each candidate mobile encoder.

2.4 Deployment

The system ships as a Progressive Web App: a portable 128-byte-per-image index, client-side Hamming ranking, and no retrieval backend (static hosting / Cloudflare). A 50k-image index is **6.4 MB** (1024-bit codes).[1] Browser inference is fp32-only—not for accuracy reasons but because the WASM execution provider cannot run int8 (`ConvInteger`) or load external-data fp16 weights (§3.4).

3 Main Results

3.1 Retrieval quality

Table 1 reports COCO 5K retrieval for the deployed 1-bit system against the float-cosine ceiling and against naive sign hashing of the frozen embedding (no learned head). The learned head recovers most of the float ceiling at 1/72 of the storage (1024 bit vs. fp32$\times$1152) and beats the headless naive baseline by +8.6 R@10 (EN) and +18.9 R@10 (KO). Korean even exceeds its own float ceiling (71.04 vs. 66.26) because the deployed head is Korean-fine-tuned, whereas the raw float tower is not.

Bit-rate. Fig. 2 sweeps code length. Quality rises steeply to 256-bit, is near-saturated by 1024-bit (R@10 80.64, EN), and gains little beyond: 2048-bit adds only +0.7 for 2$\times$ storage and 4096-bit does not improve further. 1024-bit (128 B) is the deployed sweet spot.

3.2 Multilingual

On XM3600 (36 languages, text$\rightarrow$image), the deployed server head averages **R@10 70.30**, versus a float ceiling of 74.46; the fully offline browser encoder (MiniLM-L12) averages 53.92 (Table 2). The server path leads on average, but on four low-resource, non-Latin languages—Hindi, Telugu, Thai, Maori—the offline multilingual encoder *beats* the SigLIP2 server path (e.g. Hindi 52.36 offline vs. 43.38 server), and on five it beats even the float ceiling. The multilingual text tower, not the binarization, is the bottleneck on those scripts.

3.3 Baselines

Table 3 compares the learned head against ITQ and LSH applied to the *same* frozen SigLIP2 features, in a common evaluation harness (text$\rightarrow$image). The learned head wins at every bit-rate and the margin widens dramatically as bits shrink: at 1024-bit it leads ITQ by +8.7 R@10, but at 64-bit by +55 and at 256-bit by +29—data-driven binarization matters most exactly where storage is tightest. We do *not* beat float multilingual SoTA: NLLB-CLIP's float ceiling reaches XM3600 R@10 86.06,[2] and as a server float model it is a different operating point; our claim is the 1-bit, 128-byte, backend-free regime, where a 1-bit NLLB-CLIP head reaches only 51.45.

3.4 On-device cost

Table 4 collects measured deployment costs. *Search* is interactive entirely in the client: a numpy Hamming brute-force over a 5K$\times$1024-bit index (0.64 MB) takes 1.73 ms per query and matches exact recall to within 0.8%; an exact binary FAISS index is 0.033 ms/query. *Storage* is 128 B/image (6.4 MB for 50k). *On-device image encoding* (the indexing cost) is bounded by the browser runtime: WASM-fp32 per-photo latency is 263 ms (MobileCLIP2-S0) to 1066 ms (S2) on an emulated Pixel 7. Crucially, fp32-only is forced by operator/format support, not precision: int8 codes are numerically near-lossless (EN R@10 81.04 vs. 81.24; KO 70.52 vs. 71.04), but the WASM provider lacks `ConvInteger`.

[1] *Source: paper/bits_sweep.csv (`index_MB_50k`).*

[2] *Source: paper/master_table.csv, paper/nllb_hashing.csv.*

bits	ours 1-bit	ITQ	LSH
64	**66.86**	11.92	5.64
256	**77.48**	48.52	22.88
1024	**79.92**	71.26	59.36

Table 3: COCO 5K T2I R@10, same frozen features, common harness. *Source: paper/rigor_map_bit.csv (ours), paper/rigor_baselines.csv (ITQ/LSH).*

component	cost	source
search (client numpy)	1.73 ms/query	rigor_faiss_bench
search (FAISS exact)	0.033 ms/query	rigor_faiss_bench
storage	128 B/image (6.4 MB/50k)	bits_sweep
img encode S0 (WASM)	263 ms/photo	PILLAR1_DEPLOY
img encode S2 (WASM)	1066 ms/photo	PILLAR1_DEPLOY
int8 code accuracy	81.04/70.52 R@10	precision_lc

Table 4: Measured on-device costs. Search index is 5K images; a 50k extrapolation ($\approx$0.33/15 ms) is **[VERIFY: not directly benchmarked at 50k]**. Browser text-query latency **[VERIFY: not measured in repo CSVs]**.

3.5 Encoder choice for on-device indexing

Table 5 sweeps ten candidate on-device image encoders, each head-adapted into the frozen SigLIP2 code space; the anchor (full SigLIP2 image tower) is R@10 81.24. The dominant factor is vision–language pretraining, not parameter count: MobileCLIP2-S0 (11.4 M, VL-pretrained) reaches 70.52, beating the larger image-only-SSL DINOv2-base (86.6 M, 65.0). MobileCLIP2-S2 (35.8 M) is the practical sweet spot at 76.18—near the 92.9 M SigLIP2-base (77.98) at a fraction of the size and latency (Fig. 3).

4 Analysis: Why Binarization, Not Dimension, Is the Cost

(Main-track companion — to be written.) Continuous dimension reduction is nearly free while sign binarization is the dominant low-bit loss; recoverable in part by asymmetric search.

5 Failure Modes Unique to Cross-Modal/Multilingual Hashing

(To be written.) Distillation's cross-modal binary-parity collapse; non-Latin script degradation traced to the text tower.

6 Related Work

(To be written.) OrthoHash [1] and HashCoder/CroVCA [2] establish single-loss frozen-feature hashing (BatchNorm balance); we extend to cross-modal, multilingual, and on-device, and characterize the failure modes there. **[VERIFY: phrase HashCoder cross-modal scope without quoting an unverified future-work sentence.]**

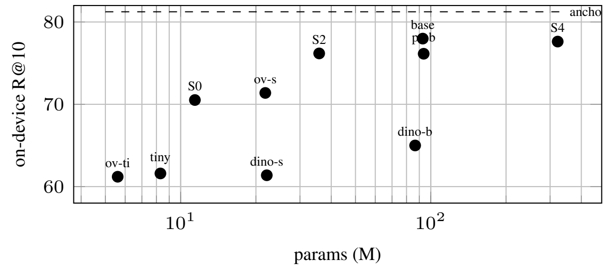

Figure 3: On-device image-indexing R@10 vs. encoder size (EN, server-side eval). VL-pretrained small encoders beat larger image-only-SSL ones. *Source: paper/image_encoders_headadapt.csv.*

encoder	params (M)	R@10
SigLIP2 image (anchor)	707.8	81.24
SigLIP2-base	92.9	77.98
MobileCLIP2-S4	321.8	77.62
MobileCLIP2-S2	35.8	76.18
PE-Core-B16	93.7	76.14
OpenVision-S16	21.8	71.38
MobileCLIP2-S0	11.4	70.52
DINOv2-base	86.6	65.0
TinyCLIP-8M	8.3	61.6
DINOv2-small	22.1	61.38
OpenVision-Ti16	5.6	61.2

Table 5: On-device encoder sweep (COCO EN R@10, head-adapted). *Source: paper/image_encoders_headadapt.csv.*

7 Conclusion

(To be written.) A deployed, backend-free 1-bit multilingual image search system; honest scope (does not beat float multilingual SoTA), measured costs, and reproducible numbers.

References

[1] Hoe, J.T.; et al. One Loss for All: Deep Hashing with a Single Cosine Similarity based Learning Objective. *NeurIPS*, 2021.

[2] Cross-View Code Alignment (HashCoder / CroVCA). arXiv:2510.27584, 2025.